CONFOEDERATIO,
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THE RUSSIAN CORE PRINCIPALITIES

A Regional Literature Review of Population and Settlement (10000BC - 1900AD).

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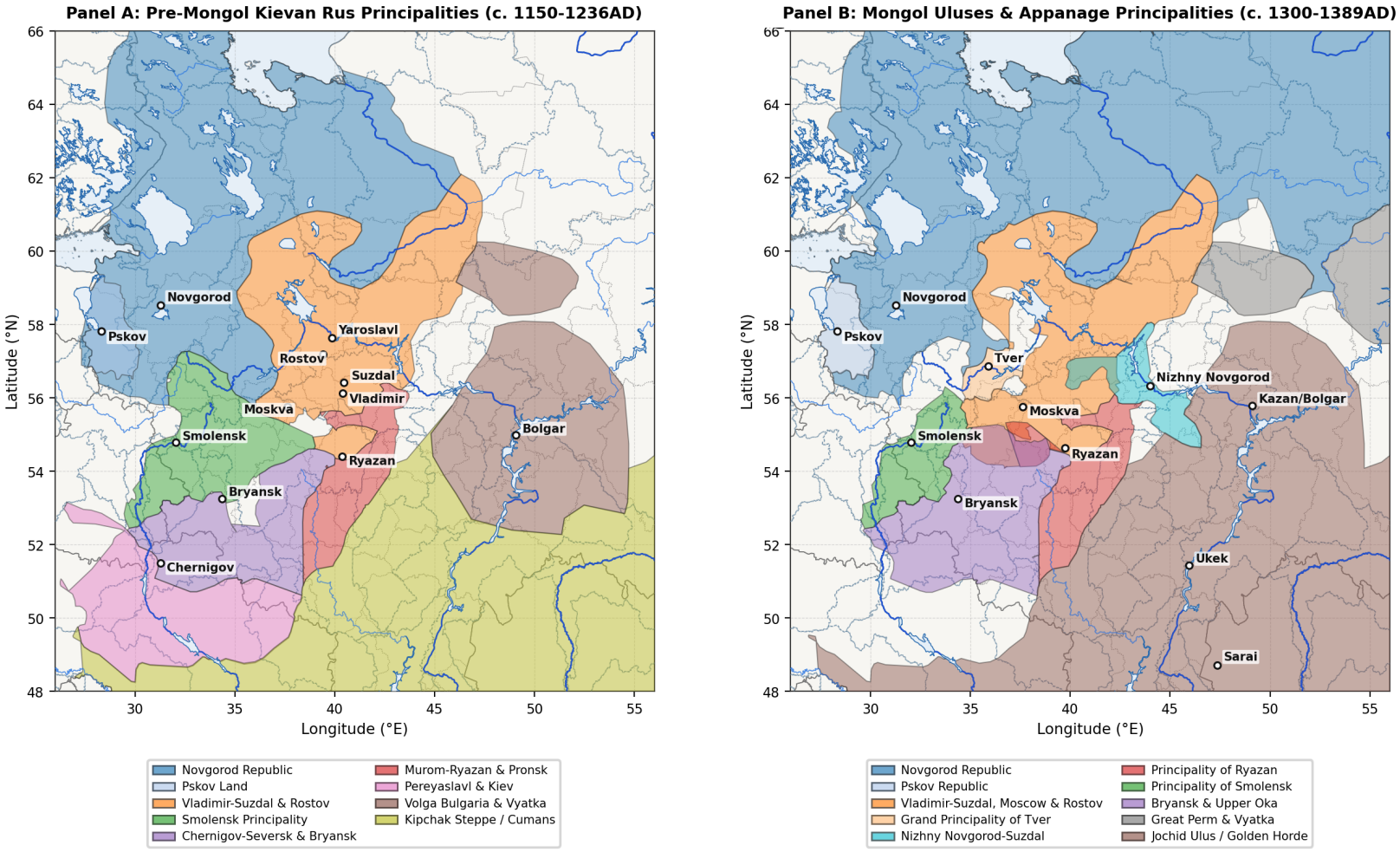


Figure 1. Regional boundaries and hydrological catchments of pre-Mongol principalities and Mongol-era polities across the Russian Core (10000BC - 1900AD).

Abstract

This review reconstructs the long-run population and settlement history of the Russian Core and the Middle Volga-Ural basin (~1.943.000 km^2) from 10000BC to 1900AD across eight historical regional divisions: Novgorod and Pskov, Vladimir-Suzdal and Rostov-Yaroslavl, Moscow and Tver, Ryazan and the Upper Don, Smolensk and the Upper Dnieper, Chernigov-Bryansk and the Upper Oka, Nizhny Novgorod, Vyatka, and Great Perm, and the Middle Volga (Volga Bulgaria and Kazan). Estimates combine excavated dwelling floor areas and hillfort networks for the prehistoric and Early Iron Age periods, urban homestead (*usadba*) and rural *selishche* surveys for Kievan Rus and Volga Bulgaria, the 1257-1259 Mongol

census (*chislo*), sixteenth- and seventeenth-century Muscovite cadastral and household books (*pistsovye* and *perepisnye knigi*), the ten Imperial Poll-Tax Revisions (1719-1858), and the First All-Russian Imperial Census of 1897. Total regional population grew from 13k foragers at 10000BC to 182k in the Bronze Age (2000BC), 610k at 1AD, and 4,67M on the eve of the Mongol invasion (1200AD), before falling to 3,85M in 1250AD. Subsequent upland watershed colonisation, southward expansion into the forest-steppe, and nineteenth-century agrarian and industrial growth raised the population to 6,49M in 1500AD, 10,19M in 1700AD, and 41,5M by 1900AD.

Background

To maintain consistent geographic boundaries across the prehistoric, Kievan Rus (882 - 1240AD), Mongol and Appanage (1240 - 1500AD), Muscovite (1500 - 1700AD), and Imperial (1700 - 1900AD) periods, the historical principalities are grouped into eight regional divisions defined by river basins and historical provincial boundaries (Table 1 and Figure 2). Because several large polities span multiple modern republics or oblasts - notably the Golden Horde, the Novgorod Republic, and the Chernigov-Seversk principalities - the estimates in this study cover the territory of the Russian Core and the Middle Volga-Ural basin (~1.943.000 km^2), excluding the Lower Volga steppe south of Ukek (51,5°N), the Ukrainian Dnieper-Desna left bank south of the Seym-Bryansk watershed (52°N), and Belarusian Polotsk-Turov-Vitebsk west of the Smolensk Dnieper gate (31°E).

Methodology

Reconstructing the population history of the East European forest, forest-steppe, and Middle Volga-Ural zones from 10000BC to 1900AD requires combining archaeological settlement evidence for prehistoric and medieval periods with fiscal and census records for the early modern and imperial eras [1][2][3]. Rather than relying on global macro-estimates or treating raw counts of excavated sites and radiocarbon dates as direct measures of population change, this study derives regional estimates from five primary classes of historical and archaeological evidence.

For prehistoric and Early Iron Age settlements between 10000BC and 850AD, population densities across the Mesolithic, Neolithic, Bronze Age, and Early Iron Age are estimated from excavated dwelling floors, hearth counts, cemetery populations, and riverine settlement spacing. These include Mesolithic peat-bog camps of the Butovo, Kunda, and Veretye traditions [4][5][6], multi-hearth timber longhouses (60 - 200 m^2, housing 15 - 40 occupants) of the Middle Holocene Lyalovo and Volosovo cultures [7][8], Bronze Age Fatyanovo, Abashevo, and Pozdnyakovo agro-pastoralist communities [9][10][11], and Early Iron Age hillfort networks (*gorodishcha*, typically 0,15 - 0,5 ha spaced every 3 - 7 km along river bluffs, supporting 50 - 150 inhabitants per fortified site) of the Dyakovo, Gorodets, Dnieper-Dvina, and Yukhnovo cultures [12][13][14].

For the urban network of Kievan Rus and Volga Bulgaria between 850 and 1450AD, population estimates follow the fortified-area and homestead (*usadba*/*dvor*) methods developed by M. N. Tikhomirov [15], A. V. Kuza [2][16], V. P. Darkevich [17], V. L. Yanin [18], and F. Sh. Khuzin [19]. Across dendrochronologically dated quarters in Veliky Novgorod (Nerevsky, Lyudin, and Slavensky ends), Pskov, Staraya Russa, Smolensk, Old Ryazan, Vladimir, and Suzdal, an urban *usadba* occupied 400 - 1.200 m^2 (averaging 600 - 800 m^2 inside timber ramparts and 900 - 1.400 m^2 in outer *posad* suburbs), resulting in 10 - 16 homesteads per net residential hectare. Applying a mean household multiplier of 5,5 - 6,5 persons per homestead (or 8 - 10 persons on large boyar compounds with dependent artisans) yields a density of 80 - 150 inhabitants per walled hectare [2][17].

Rural population estimates between 900 and 1500AD are grounded in total-coverage field walking and

geophysical surveys of unfortified villages (*selishcha*) in the Suzdal Opolye [20][21][22], the Lake Ilmen and Volkhov basin [23], the Moskva-Radonezh upland watersheds [24][25], the Upper Don valley [26], and the Western Trans-Kama (*Zakamye*) agricultural core of Volga Bulgaria [27][28]. Whereas a typical northern forest hamlet (*derevnya* of 1 - 3 homesteads) covered 0,3 - 1,2 ha (15 - 30 inhabitants), large Opolye and Bulgar chernozem villages (*sela*) covered 5 - 19 ha and housed 100 - 350 inhabitants [19][21].

For the Mongol and early modern periods between 1257 and 1700AD, estimates are cross-checked against the Empire-wide Mongol census (*chislo*) conducted under Grand Qan Möngke and Berke in 1257 - 1259 [29][30][31], the late fifteenth- and sixteenth-century Novgorodian and Muscovite *pistsovye knigi* (land cadastres recording peasant *dvory* and *obzha*/*sokha* tax units) [32][33][34], and the 1646 and 1678 Empire-wide household census books (*perepisnye knigi*) analysed by Ya. E. Vodarsky [1][3][35]. From the reign of Peter I to the end of the nineteenth century (1719 - 1900AD), provincial population totals are anchored to the ten Imperial Poll-Tax Revisions (*revizskie skazki*, 1719 - 1858) compiled by V. M. Kabuzan [36][37][38][39], nineteenth-century administrative statistics synthesised by A. G. Rashin and B. N. Mironov [40][41][42][43], and the First All-Russian Imperial Census of 28 January 1897 (*Pervaya vseobshchaya perepis naseleniya Rossiyskoy Imperii*) edited by N. A. Troynitsky [44][45], carried forward to 1900AD.

Regional Division	Constituent Medieval Polities	Kievan Span	Mongol / Appanage	Corresponding Imperial Provinces & Area
Novgorod, Pskov, St. Petersburg & N. Dvina	Novgorod Republic, Pskov Republic, Zavolochye	882 - 1240AD	1240 - 1478/ 1510AD	St. Petersburg, Novgorod, Pskov, Olonets, Arkhangelsk (680.000 km^2)
Vladimir-Suzdal, Rostov-Yaroslavl & Beloozero	Vladimir-Suzdal, Rostov, Yaroslavl, Beloozero	988 - 1238AD	1238 - 1474AD	Vladimir, Yaroslavl, Kostroma, Vologda (285.000 km^2)
Moscow & Tver	Grand Principalities of Moscow and Tver	1147 - 1238AD	1247 - 1485AD	Moscow, Tver (92.000 km^2)
Ryazan, Murom, Pronsk & Upper Don	Murom-Ryazan, Pronsk, Yelets, Lipetsk	1097 - 1237AD	1237 - 1521AD	Ryazan, N. Tambov, N. Voronezh / Upper Don (105.000 km^2)
Smolensk, Toropets & Upper Dnieper	Smolensk, Toropets, Vyazma, Dorogobuzh, Mstislavl	1054 - 1239AD	1239 - 1404/ 1514AD	Smolensk, E. Toropets / Bely (78.000 km^2)
Chernigov-Bryansk, Kursk & Upper Oka	Chernigov-Kursk, Bryansk, Upper Oka Principalities	1024 - 1239AD	1239 - 1503AD	Kursk, Orel, Tula, Kaluga, Bryansk (108.000 km^2)
Nizhny Novgorod, Vyatka & Great Perm	Nizhny Novgorod-Gorodets, Vyatka Land, Great Perm	1152 - 1239AD	1239 - 1489/ 1505AD	Nizhny Novgorod, Vyatka, Perm (420.000 km^2)
Middle Volga (Bulgar, Kazan, Mukhshi, Kasimov)	Volga Bulgaria, Bulgar and Mukhshi Uluses, Kazan, Kasimov	922 - 1236AD	1236 - 1552AD	Kazan, Simbirsk, Penza, Kasimov/ Temnikov (175.000 km^2)

Table 1. Regional principality divisions, constituent medieval polities, historical date spans, and corresponding imperial provinces of the Russian Core.

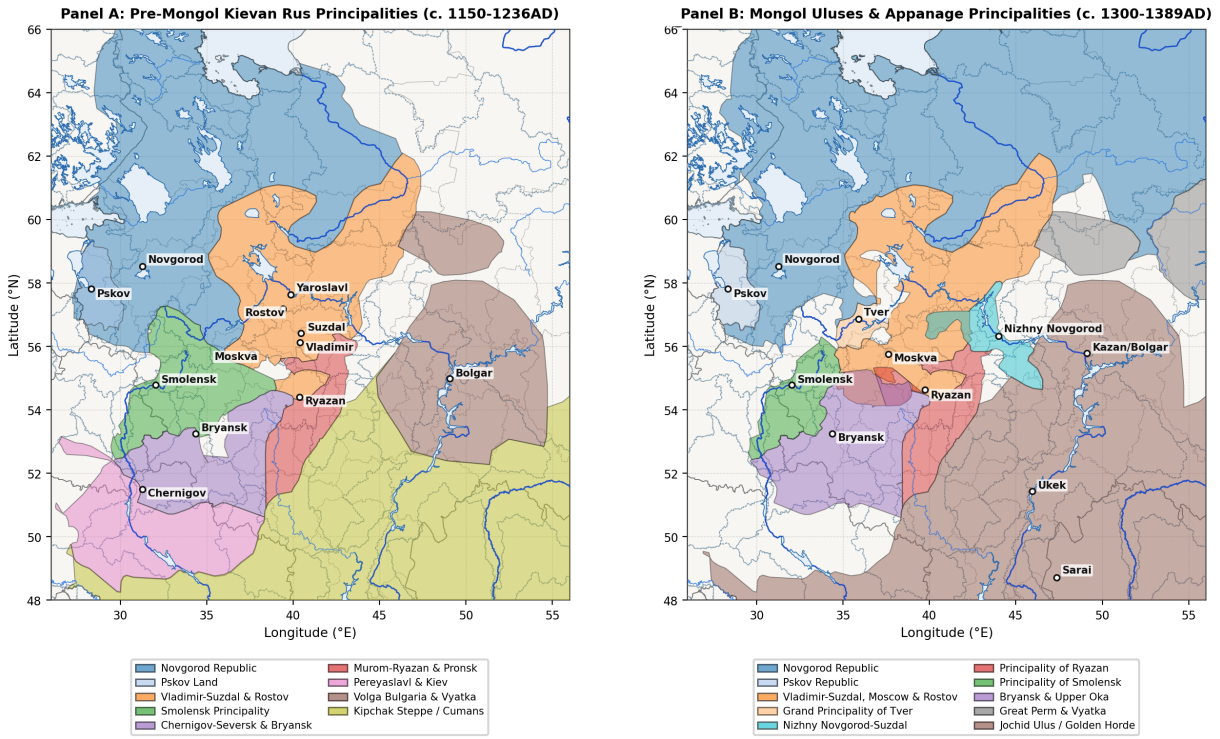


Figure 2. Territorial boundaries and hydrological basins of the pre-Mongol principalities, Volga Bulgaria, and Mongol-era polities across the Russian Core.

Results

Prehistoric and Early Iron Age Settlement (10000BC - 850AD)

Mesolithic and Pit-Comb Neolithic Foragers and Fishers

Human recolonisation of the East European forest zone began during the Younger Dryas-Preboreal transition (10000 - 8000BC) as mobile reindeer- and elk-hunting bands followed periglacial river valleys northward [4][5]. Across the Volga-Oka interfluve and the Upper Dnieper-Desna basin, the Butovo and Resseta tanged-point cultures (8000 - 5500BC) established seasonal base camps on sandy lacustrine terraces and peat bogs such as Stanovoye 4, Ivanovskoye 7, and Zamostje 2 [5][6]. In the north-western lake basins (Lakes Ilmen, Lacha, Vozhe, Kubenskoye, and the Sukhona basin), Kunda and Veretye forager-fishers occupied log-built shoreline settlements such as Nizhneye Veretye I (6800 - 6000BC), where waterlogged organic preservation reveals specialised bone harpoons, weir traps, and birch-bark floats [4]. Because Mesolithic subsistence depended on seasonal ungulate migrations and lacustrine fish runs, regional population densities remained low (0,006 - 0,025 persons/km^2), with the entire Russian Core and Middle Volga-Ural territory increasing from roughly 13k inhabitants in 10000BC to 44,5k by 6000BC [6].

During the Atlantic climatic optimum (5500 - 2800BC), the spread of Upper Volga early ceramic traditions (5500 - 4200BC), the Lyalovo pit-comb ware culture (4200 - 3000BC), and the Eneolithic Volosovo culture (3200 - 2000BC) supported a much denser network of semi-sedentary fishing and hunting villages along lake outlets across the Volga-Oka, Klyazma, and Kama basins [7][8]. At the Sakhtysh I-VIII peat-bog complex near Ivanovo and multi-layer sites on lakes Nero and Pleshcheyevo, Volosovo communities constructed rectangular semi-subterranean timber longhouses measuring 60 - 200 m^2 with 3 - 6 axial hearths, each accommodating an extended household of 15 - 40 persons, alongside structured cemeteries

exceeding 100 burials [7][8][10]. Across all eight regional divisions, the total population rose from 86k in 4000BC to 122k by 3000BC.

Bronze Age Agro-Pastoralists and Iron Age Hillfort Networks

Around 2800 - 2000BC, the arrival of Corded Ware groups of the Fatyanovo-Balanovo culture introduced domesticated cattle, sheep, goats, pigs, and wheeled vehicles into the Upper and Middle Volga forest zone [9][10]. Ancient DNA analyses from Fatyanovo cemeteries in Yaroslavl, Ivanovo, Moscow, and Chuvashia (including Volosovo-Danilovsky, featuring over 100 graves) confirm that Fatyanovo pastoralists carried steppe-derived ancestry mixed with Central European Early Farmers and coexisted for several centuries with indigenous Volosovo hunter-fishers before genetic and cultural integration [9][10]. Subsequent Middle and Late Bronze Age societies - the Abashevo (2200 - 1700BC), Seima-Turbino (2100 - 1800BC), Pozdnyakovo (1800 - 1200BC), and Prikazan (1600 - 900BC) cultures - expanded copper-tin metallurgy along the Oka, Klyazma, Kostroma, and Kama rivers, as documented at the Starshe-Nikitinsky cemetery and the Govyadinovo settlement [11][46]. Total regional population reached 182k in 2000BC and 278k by 1000BC.

The transition to the Early Iron Age (800BC - 500AD) brought local bog-iron smelting, stockbreeding (dominated by horses, cattle, and pigs), and forest swidden millet-barley cultivation [12][13][14]. Across the Upper Volga, Moskva, and Oka basins, the Finno-Ugric Dyakovo culture (700BC - 500AD) constructed hundreds of promontory hillforts (*gorodishcha*) protected by concentric ditches and log palisades, while the related Gorodets culture occupied the Middle Oka and Moksha-Sura forest-steppe, the Dnieper-Dvina and Yukhnovo cultures occupied the Smolensk and Bryansk-Desna basins, and the Ananyino (800 - 300BC), Pyanobor (200BC - 300AD), and Glyadenovo cultures flourished along the Kama and Vyatka valleys [12][47]. N. A. Krenke's excavations at Dyakovo Gorodishche, Kuntsevo, and Borsheva in the Moskva basin show that Iron Age hillforts averaged 0,15 - 0,5 ha of enclosed habitation, spaced at 3 - 7 km intervals along main rivers and housing 50 - 150 inhabitants (8 - 20 timber dwellings), surrounded by small satellite hamlets (*selishcha*) [12][13][48]. By 1AD, the total population of the eight regions reached 610k, expanding to 920k in 500AD and 1,39M by 800AD as Early Slavic groups (Ilmen Slovenes, Krivichi, Vyatichi, and Severians) and Early Bulgars entered the forest and forest-steppe zones [49][50].

Kievan Rus Principalities (850 - 1240AD)

Novgorod and Pskov

Between the ninth and eleventh centuries, the north-western lake basin developed around the Baltic-Volga and "Varangians to the Greeks" river trade corridors [23][49]. Early proto-urban trading and craft centres at Staraya Ladoga (c. 753AD), Rurikovo Gorodishche at the source of the Volkhov (ninth to tenth centuries, covering 4 - 7 ha), and the *sopki* high-mound cemetery network around the Lake Ilmen Poozerye gave rise by the mid-tenth century to the three initial urban nuclei (*kontsy*: Nerevsky, Slavensky, and Lyudin) of Veliky Novgorod [18][23]. By 1200AD, the outer ramparts (*Okolny Gorod*) of Veliky Novgorod enclosed roughly 240 ha, containing 4.000 - 5.000 fenced courtyard homesteads (*usadby* of 600 - 1.200 m^2 excavated at the Troitsky and Nerevsky trenches) and supporting an urban population of 25k - 30k [2][15][18].

In the western Lake Peipus-Velikaya basin, Pskov expanded from its 3 ha Krom promontory fortress (eighth to tenth centuries) across the Dovmontov Gorod and Middle and Outer Town ramparts (~105 ha by the fourteenth century) to 12k - 16k inhabitants, complemented by Izborsk, Ostrov, Opochka, and Gdov [51][52]. Other major eleventh- and twelfth-century Novgorodian towns included Staraya Russa (8k - 10k, a major salt-boiling centre south of Lake Ilmen), Torzhok (5k - 7k on the Tvertsa portage), Ladoga (3k - 4k), Velikiye Luki, and Koporye [2]. Across the agricultural core of the Five Pyatinas (Vodskaya, Obonezhskaya, Bezhetskaya, Derevskaya, and Shelonskaya) and the Northern Dvina *Zavolochye* fur-trapping colonies, the total population of the Novgorod-Pskov region rose from 285k in 900AD to 380k

in 1000AD and 620k on the eve of the Mongol invasion (1200AD) [32].

Vladimir-Suzdal, Rostov-Yaroslavl, and Beloozero

In North-Eastern Rus, Slavic agricultural colonisation by Krivichi and Ilmen Slovene settlers merged with the indigenous Finno-Ugric Merya and Ves populations between the tenth and twelfth centuries [22][53][54]. The demographic heartland of the principality was the Suzdal Opolye (~4.500 km^2 of fertile, well-drained dark carbonate loess-like soils between Suzdal, Vladimir, Yuriev-Polsky, and Pereyaslavl-Zalessky). Over twenty years of systematic field surveys and geophysical magnetometry led by N. A. Makarov, I. N. Modin, and S. V. Shpolyanskiy have documented nearly 400 medieval rural settlements (*selishcha*) in the Opolye alone [20][21]. Roughly 25 - 30% of twelfth- and early thirteenth-century Opolye villages (such as Shekshovo-2/9, Gnezdilovo, Kideksha, and Ves) were large multi-homestead settlements covering 5 - 19 ha (100 - 350 inhabitants each), yielding local rural densities of 18 - 28 persons/km^2 within the Opolye loess islands [20][21].

Under Princes Yuri Dolgoruky (1125 - 1157), Andrey Bogolyubsky (1157 - 1174), and Vsevolod the Big Nest (1176 - 1212), a dense urban hierarchy took shape across the Klyazma-Upper Volga basin [54][55]. By 1200 - 1236AD, the fortified capital of Vladimir-on-Klyazma enclosed 145 ha across its Pecherny Gorod, Novy Gorod, and Vetchany Gorod ramparts, housing 20k - 25k inhabitants; Rostov Veliky (65 ha), Suzdal (49 ha), and Pereyaslavl-Zalessky (42 ha) each supported 8k - 12k residents; and Yaroslavl, Yuriev-Polsky, Kostroma, Galich-Mersky, Starodub-on-Klyazma, Gorodets-Radilov (60 ha), Uglich, and Dmitrov housed 3k - 7k each [2][16][56]. Further north along Lake Beloye and the Sheksna River, the town of Beloozero (Kema/Krokhino and the fourteenth-century Belozersk ramparts) anchored a network of Finno-Slavic settlements documented by A. V. Kudryashov [57]. In total, the Vladimir-Suzdal, Rostov-Yaroslavl, and Beloozero region expanded from 310k in 900AD and 430k in 1000AD to 920k by 1200AD [56].

Smolensk, Chernigov-Bryansk, the Upper Oka, and Murom-Ryazan

Along the Upper Dnieper and Western Dvina portages, tenth-century urbanism centred on the Gnezdovo archaeological complex (12 km west of Smolensk), comprising two fortified settlements (17 - 20 ha of habitation) and over 4.500 barrows (*kurgany*), with an estimated population of 1,5k - 2,5k permanent Scandinavian, Slavic, and Baltic residents [58][59]. In the eleventh and twelfth centuries, princely administration shifted to Smolensk, whose 115 ha fortified enclosure and dense *posad* artisan quarters along the Dnieper supported 20k - 30k inhabitants on the eve of the 1230 famine and pestilence chronicle entry, supplemented by Toropets, Roslavl, Mstislavl, Dorogobuzh, Vyazma, and Yelnya [2][15][60]. Across its 56 tributary *pogosty* recorded in the 1136 Statutory Charter of Prince Rostislav Mstislavich, the Smolensk principality grew from 210k in 1000AD to 430k in 1200AD [60].

South and east of Smolensk, the Russian sector of the Chernigov-Seversk and Upper Oka lands encompassed the Vyatichi tribal territory along the Upper Oka, Zhizdra, Upa, and Zusha rivers together with the Desna-Bryansk and Posemye (Kursk-Rylsk) basins [50][61]. As shown in Figure 3, A. V. Kuza's survey of tenth- to thirteenth-century fortified settlements (*gorodishcha*) demonstrates that the Chernigov-Seversk and Upper Oka region possessed one of the densest networks of small fortified administrative and refuge towns (<2,5 ha) in Rus, including Kozelsk, Karachev, Bryansk (*Debryansk*), Mtsensk, Novosil, Tarusa, Vorotynsk, Odoev, Belyov, Mezetsk, Mosalsk, Serensk (famous for its twelfth-century jewellery moulds), Dedoslavl, Spash-Gorodok, and Kursk [2][16][61]. Combined with the major urban centres of the Desna-Seym basin, the Russian sector of the Chernigov-Bryansk and Upper Oka region rose from 310k in 1000AD to 580k in 1200AD.

In the Middle Oka and Upper Don forest-steppe, the Murom-Ryazan and Pronsk principalities expanded rapidly during the twelfth century [17]. Excavations by V. P. Darkevich and G. V. Borisevich at Old Ryazan

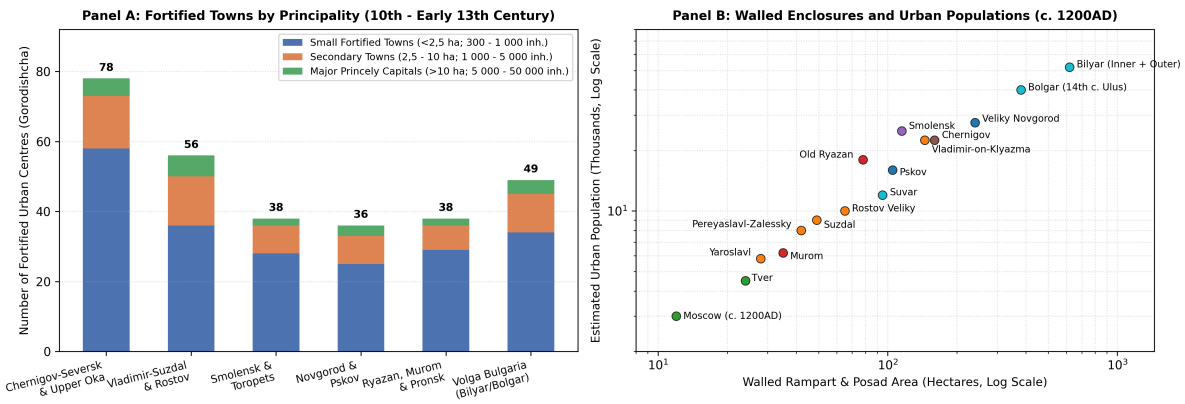


Figure 3. Urban hierarchy, fortified enclosure areas, and population estimates across the Russian Core and Volga Bulgaria (c. 1000 - 1236AD).

(*Staraya Ryazan*, near Spassk-Ryazansky) show that its Northern (8 ha), Southern (60 ha), and suburban *posad* enclosures (~78 ha total) contained 1.500 - 2.500 timber homesteads and craft workshops, supporting 15k - 20k inhabitants before its destruction by Batu Khan in December 1237 [2][17]. Along with Murom (35 ha, 6k - 8k), Pronsk (4k - 6k), Pereyaslavl-Ryazansky, Kolomna, Rostislavl, and Belgorod-Ryazansky, and over 250 twelfth- to fourteenth-century rural *selishcha* surveyed along the Upper Don and Nepryadva rivers [26], the Ryazan, Murom, and Upper Don region reached 520k inhabitants by 1200AD (Table 2).

Urban Centre	Region	Walled & Posad Area (ha)	Estimated Homesteads	Estimated Population
Bilyar (Volga Bulgaria)	Middle Volga	620 (800 w/ suburbs)	7.500 - 9.500	45k - 60k (c. 1200)
Bolgar (Golden Horde Ulus)	Middle Volga	380 (14th c. rampart)	5.500 - 7.000	35k - 45k (c. 1340)
Veliky Novgorod	Novgorod and Pskov	240 (Okolny Gorod)	4.200 - 5.000	25k - 30k (c. 1200)
Smolensk	Smolensk and Upper Dnieper	115 (Detinets + Posad)	3.500 - 4.500	22k - 28k (c. 1200)
Vladimir-on-Klyazma	Vladimir-Suzdal	145 (3 rampart rings)	3.200 - 4.000	20k - 25k (c. 1200)
Chernigov (Desna Core)	Chernigov-Bryansk	160 (Detinets + Okolny)	3.200 - 4.000	20k - 25k (c. 1200)
Old Ryazan (Staraya Ryazan)	Ryazan and Upper Don	78 (North + South + Posad)	2.200 - 3.000	15k - 20k (c. 1236)
Pskov	Novgorod and Pskov	105 (Krom + Outer Wall)	2.200 - 2.800	14k - 18k (c. 1350)
Suvar (Volga Bulgaria)	Middle Volga	95 (Walled Shahristan)	1.800 - 2.200	10k - 14k (c. 1200)
Rostov Veliky	Rostov-Yaroslavl	65 (Rampart enclosure)	1.500 - 1.900	9k - 12k (c. 1200)
Suzdal	Vladimir-Suzdal	49 (Kremlin + Okolny)	1.300 - 1.600	8k - 10k (c. 1200)
Pereyaslavl-Zalessky	Vladimir-Suzdal	42 (1152 Earth Rampart)	1.100 - 1.400	7k - 9k (c. 1200)

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Urban Centre	Region	Walled & Posad Area (ha)	Estimated Homesteads	Estimated Population
Staraya Russa	Novgorod and Pskov	38 (Saline Posad)	1.200 - 1.500	8k - 10k (c. 1200)
Murom	Ryazan and Murom	35 (Kremlin + Posad)	950 - 1.200	6k - 7,5k (c. 1200)
Yaroslavl	Rostov-Yaroslavl	28 (13th c. Ramparts)	850 - 1.100	5k - 7k (c. 1200)
Tver	Moscow and Tver	24 (1200)/85 (1400)	750/2.200	4,5k (1200)/14k (1400)
Moscow (1200/ 1450AD)	Moscow and Tver	12 (1200)/190 (1450)	450/6.500	3k (1200)/40k (1450)

Table 2. Fortified enclosure areas, estimated homestead counts, and populations of principal medieval cities (1200 - 1450AD).

Mongol Census, Appanages, and Volga Khanates (1240 - 1500AD)

The 1237-1240 Invasion, the 1257-1259 Census, and Appanage Settlement

Batu Khan's winter campaign of 1237 - 1238 against Ryazan, Kolomna, Moscow, Vladimir, Suzdal, Rostov, Yaroslavl, Tver, Torzhok, and Kozelsk - preceded by Subutai's 1236 conquest of Volga Bulgaria and followed by the 1239 campaigns against Murom, Gorodets, and Chernigov - inflicted severe demographic losses on the fortified urban centres and exposed forest-steppe valleys [31][62][63]. Out of 74 Russian towns known from twelfth- and early thirteenth-century archaeological layers, 49 were burned during 1237 - 1240, and 14 (including Old Ryazan and Izheslavl) never recovered their prior urban status [2][17]. Between 1200 and 1250AD, total population across the eight regional divisions fell by roughly 18%, from 4,67M to 3,85M, with the steepest declines occurring in Ryazan (-28,8%), Chernigov-Upper Oka (-25,9%), Volga Bulgaria (-22,4%), and Vladimir-Suzdal (-19,6%), whereas uninvaded Novgorod and Pskov experienced only minor border losses (-4,8%) [56][62].

To organise systematic taxation (*vykhod/dan'*) and military conscription across the Jochid Ulus, Grand Qan Möngke and Berke dispatched imperial census officials (*chislenniki*) in 1257 - 1259 to register households in Suzdalia, Ryazan, Murom, and (with Alexander Nevsky's military enforcement in 1259) Veliky Novgorod [3][29][31]. As analysed by G. Vernadsky [30] and T. T. Allsen [29], the Mongol administrative system grouped registered tax-paying peasant households into decimal units of ten, hundred, thousand (*tysyacha*), and ten thousand (*tumen/t'ma*). The 27 to 43 *t'my* cited in fourteenth-century princely treaties and chronicle entries across North-Eastern Rus and Novgorod - where a fiscal *t'ma* in wooded agricultural zones corresponded to 10.000 - 15.000 tax-paying households (60k - 90k inhabitants) rather than a full steppe cavalry corps - align closely with an aggregate post-invasion population of 3,85M in 1250AD, recovering to 4,33M in 1300AD and 4,84M on the eve of the Black Death (1350AD) [29][30][54].

Between the late thirteenth and fifteenth centuries, archaeological surveys document a profound reorganisation of rural settlement across the Moscow, Tver, and Vladimir-Suzdal principalities known in Russian historiography as the "ascent to the hills" (*vzlet na kholmy*) or internal watershed colonisation [21][24][25]. Rather than remaining concentrated along major river floodplains vulnerable to Golden Horde punitive raids (*rati* such as Dyuden's campaign of 1293 or Tokhtamysh's sack of 1382), peasant settlers cleared upland morainic loams using iron-shod *sokha* ploughs and three-field crop rotation (winter

rye, spring oats, and fallow). S. Z. Chernov's total archaeological surveys of the Radonezh, Zvenigorod, Vorya, and Pereyaslavl *volosti* demonstrate that between 1280 and 1450AD hundreds of small 1 - 3 household clearing hamlets (*pochinki* and *derevni*, 0,2 - 0,8 ha) multiplied across interfluvial hills under the patronage of the Grand Princes of Moscow and monastic estates such as the Trinity-Sergius Lavra [24][25]. Simultaneously, the Grand Principality of Tver (including Kashin, Mikulin, Zubtsov, Kholm, and Staritsa) and the Principality of Nizhny Novgorod-Suzdal (founded 1341 at the Volga-Oka confluence) emerged as major demographic and commercial rivals to Moscow [54][64]. Although the second wave of the Black Death (1352 - 1353, which killed Grand Prince Simeon the Proud and Metropolitan Theognostus, followed by recurrent plague outbreaks in 1364 - 1366, 1389, and 1417 - 1427) temporarily reduced total population from 4,84M in 1350AD to 4,49M in 1400AD, rapid upland colonisation raised the Moscow-Tver region from 330k in 1250AD to 830k in 1450AD and 1,08M by 1500AD [33][65].

Volga Bulgaria, the Golden Horde Uluses, and the Kazan, Kasimov, and Temnikov Polities

In the Middle Volga-Kama basin, pre-Mongol Volga Bulgaria (922 - 1236AD) was one of the most urbanised states in Eastern Europe [19][27]. F. Sh. Khuzin's and K. A. Rudenko's syntheses of over 170 Bulgar fortified towns (*gorodishcha*) and more than 700 unfortified villages (*selishcha*) show that the twelfth-century capital of Bilyar (*Velikiy Gorod* in Russian chronicles) encompassed 620 ha inside its triple earthen ramparts (800 ha including outer suburbs), with brick-heated residences, glass and metallurgical workshops, and a peak population of 45k - 60k [19][27]. Other major Bulgar towns included Suvar (95 ha, 10k - 14k), Bolgar (25 - 30 ha in the twelfth century), Oshel, Juketau (*Zhukotin*), Kashan, Muromsky Gorodok (150 ha on the Samara Bend), and Kazan (founded c. 1005AD) [19]. Across the fertile chernozem plains of Western Trans-Kama (*Zakamye*: the Cheremshan, Sheshma, Aktay, and Maly Cheremshan valleys), pre-Mongol Volga Bulgaria supported 850k inhabitants by 1200AD [27].

After the 1236 Mongol conquest destroyed Bilyar, Oshel, and Suvar, the region was reorganised as the Bulgar Ulus and the southern Ulus of Mukhshi (Narovchat) within the Golden Horde [28][66][67]. Under Mengu-Timur and Uzbeg Khan (1313 - 1341), Bolgar expanded into a 380 ha walled metropolis (35k - 45k inhabitants) serving as the northern summer residence and mint of the Jochid khans [28], while in the Moksha-Sura forest-steppe (modern Penza Oblast and Mordovia) the city of Mukhshi (Narovchat, exceeding 150 ha, with brick mosques, mausoleums, baths, and a silver mint established in 1313) anchored a dense network of Mordvin, Burtas, and Tatar settlements [66][67]. East of the Kama in the Ik-Belaya interfluve, Tatar-Muslim agricultural and pastoral communities also expanded during the thirteenth and fourteenth centuries [68].

However, the Golden Horde's "Great Troubles" (*Velikaya Zamknutnya*, 1359 - 1380), Bulat-Timur's 1361 seizure of Bolgar, Muscovite river raids (1376, 1399), Timur's (Tamerlane's) 1395 campaign, and nomadic Nogai incursions triggered a decisive northward migration from the exposed Southern Zakamye steppe borderland into the forested Predkamye (north of the Kama: the Kazanka, Mesha, Ashit, Ilet, and Sviyaga basins) [27][34]. This demographic shift laid the foundation of the Khanate of Kazan (1438 - 1552AD), administered through five territorial *darugas* (Alat, Arsk, Galich, Zyuri, and Nogai) and populated by Kazan Tatars, Chuvash, Mari (Cheremis), Udmurts (Votyaks), and Mordvins (860k across the Middle Volga region in 1500AD) [34][69]. On the Oka-Moksha frontier between Ryazan and Kazan, the Kasimov Khanate (founded c. 1452 at Gorodets-Meshchersky) and the Temnikov Principality (ruled by Tatar princes of the Bekhanid line over Mordvin *belyaki* tribute districts) formed a densely settled military buffer zone documented in sixteenth- and seventeenth-century service registers [70][71]. Meanwhile, along the Vyatka and Upper Kama rivers, Russian colonists from Novgorod, Ustyug, and Galich established the autonomous Vyatka Land (centred on Khlynov, Kotelnich, and Orlov, together with the Arsk and Karino Tatar princes settled along the Cheptsa River; annexed by Ivan III in 1489) and the Principality of Great Perm (centred on Cherdyn, Pokcha, and Iskor on the Kolva-Vishera-Kama confluence, arising from the medieval

Komi-Permyak Lomovatovo and Rodanovo cultures; subordinated in 1472 - 1505) [47][72][73][74][75].

Muscovite Cadastral Registers and the 1678 Household Census (1500 - 1700AD)

Following the unification of the northern principalities under Ivan III (1462 - 1505) and Vasily III (1505 - 1533), the Muscovite state compiled land cadastres (*pistsovye knigi*) that provide household-by-household inventories of towns and rural *pogosty* [3][32][33]. In the Novgorod Five Pyatinas (1495 - 1505), A. L. Shapiro and A. I. Kopanev's analysis of surviving cadastral books documents approximately 65.000 - 70.000 peasant homesteads (*dvory*) assessed in *obzha* plough-units (1 *obzha* = 1 man with 1 horse ploughing ~15 *desyatiny* across three fields), corresponding to 500k - 550k inhabitants in the Five Pyatinas proper and 810k across the entire Novgorod-Pskov-Northern Dvina region in 1500AD, rising to 940k by 1550AD [32][33]. Across the Zamoskovny Krai (Moscow, Tver, Vladimir, Suzdal, Pereyaslavl, Rostov, Yaroslavl, Kostroma, and Beloozero), early sixteenth-century cadastres indicate 2,57M inhabitants in 1500AD (1,49M in Vladimir-Suzdal and Rostov-Yaroslavl, and 1,08M in Moscow-Tver), expanding to 3,09M by mid-century (1550AD) as three-field arable farming reached its highest pre-industrial extent in the non-chernozem forest zone [1][33]. In the newly annexed Kazan *uezd*, the 1565 - 1568 *pistsovye knigi* of Nikita Borisov and Dmitry Kikin register over 700 Russian, Tatar, and Chuvash settlements across the Kazanka, Sviyaga, and Kama *darugas* [34][69].

During the late sixteenth and early seventeenth centuries, the northern and western regions suffered two severe demographic crises: first, the Livonian War (1558 - 1583), the Oprichnina (1565 - 1572, including Ivan IV's 1570 sack of Novgorod), and the 1569 - 1571 famine and plague, which left 60 - 85% of registered *obzhi* in the Novgorod Pyatinas and Moscow *uezd* recorded as abandoned waste (*pustoshi*) in the 1580s cadastres [32][76]; and second, the Time of Troubles (*Smuta*, 1598 - 1613), triggered by the 1601 - 1603 volcanic-winter famine and Polish-Lithuanian and Swedish military intervention [1][65]. Across all eight regional divisions, total population declined from 7,65M in 1550AD to 6,75M in 1600AD.

Recovery during the seventeenth century was accompanied by a decisive geographical redistribution of population southward across the Belgorod and Simbirsk Fortified Lines (*Zasechnaya cherta*, constructed 1635 - 1654), documented in the 1646 and 1678 Empire-wide household census books (*perepisnye knigi*) analysed by Ya. E. Vodarsky [1][3][35]. Because seventeenth-century peasant families frequently combined married brothers or adult sons inside a single fenced courtyard to reduce household (*dvor*) taxation, Vodarsky established that the average rural household size in 1678 had risen to 6,5 - 8,5 persons (3,5 - 4,2 males per *dvor*). As illustrated in Figure 4, between 1500 and 1678/1700AD population in the old north-western forest core barely surpassed its mid-sixteenth-century level (1,01M in 1700AD), whereas the forest-steppe provinces along the Upper Oka, Kursk, Orel, and Tula (doubling from 650k in 1500AD to 1,35M in 1700AD), the Ryazan and Upper Don black-earth frontier (rising from 560k to 1,05M), the Middle Volga (rising from 860k to 1,18M), and the Vyatka-Kama-Perm lands (rising from 610k to 915k) expanded rapidly, bringing the aggregate population of the eight regions to 10,19M (10,19M) by 1700AD [35].

Imperial Tax Revisions and the 1897 Census (1700 - 1900AD)

The transition from household taxation to the individual male soul tax (*podushnaya podat'*) under Peter I (1718 - 1724) inaugurated a regular series of ten Imperial Poll-Tax Revisions (*revizii*/*revizskie skazki*) between 1719 and 1858, compiled and cross-checked with parish vital registers by V. M. Kabuzan [36][37][38][39], followed by late nineteenth-century zemstvo statistics and the First All-Russian Imperial Census of 28 January 1897 [40][41][44]. Across the two centuries between 1700 and 1900AD, the total population of the eight historical regions quadrupled from 10,19M (10,19M) to 41,5M (41,5M), driven by three major regional processes (Figure 5).

In the north-western provinces, Peter I's foundation of Saint Petersburg in the Neva delta in 1703 (within

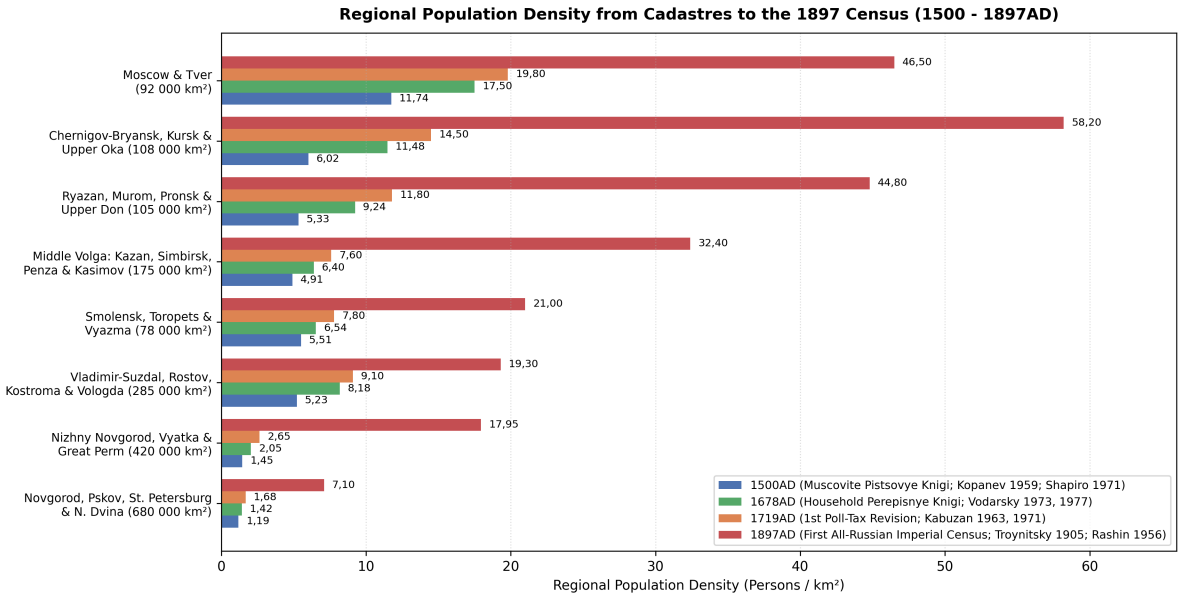


Figure 4. Regional population densities across the eight principal regions at four fiscal and census benchmarks between 1500AD and 1897AD.

the historical Ingrian and Vodskaya Pyatina territory of Novgorod) created a rapidly growing imperial capital and maritime port that reached ~95k inhabitants by 1750, 220k in 1800, 490k in 1850, and 1,26M in the 1897 Census (2,11M across Saint Petersburg Guberniya) [40][44]. Combined with Novgorod (1,37M in 1897), Pskov (1,12M), Olonets (364,2k), and the Northern Dvina and Arkhangelsk districts, the north-western region rose from 1,01M in 1700AD to 1,95M in 1800AD and 4,95M by 1900AD [37][44].

Throughout the eighteenth and first half of the nineteenth centuries (First through Tenth Revisions, 1719 - 1858), the fertile Central Black Earth (*Chernozem*) forest-steppe provinces south of the Oka River experienced the highest rural natural increase and peasant settlement densities in the Russian core [37][42][43]. Across Kursk Guberniya (2,37M in 1897), Orel Guberniya with Bryansk and Karachev (2,03M), Tula Guberniya (1,42M), and Kaluga Guberniya (1,13M), the regional population climbed from 1,35M in 1700AD to 2,95M in 1800AD and 6,45M in 1900AD (58,2 persons/km^2 in 1897) [40][44]. Similarly, in Ryazan Guberniya (1,8M in 1897) and the northern Upper Don *uezds* of Tambov and Voronezh Guberniyas (Lipetsk, Yelets, Lebedyan, Dankov, Usman, Shatsk, and Morshansk), population rose from 1,05M in 1700AD to 2,25M in 1800AD and 4,85M by 1900AD [43][44].

In the old Zamoskovny core, thin podzolic soils encouraged seasonal craft migration (*otkhodnichestvo*), proto-industrial linen and cotton manufacturing (centred on Moscow, Ivanovo-Voznesensk, Shuya, Orekhovo-Zuyevo, Yaroslavl, and Kostroma), and post-1861 railway urbanisation [39][41]. By the 1897 Census, Moscow Guberniya reached 2,43M (with Moscow city at 1,04M) and Tver Guberniya reached 1,77M, bringing the Moscow-Tver division to 4,42M in 1900AD; simultaneously, the Vladimir-Suzdal and Rostov-Yaroslavl division - spanning Vladimir (1,52M), Kostroma (1,39M), Vologda (1,34M), and Yaroslavl (1,07M) together with the Belozersk and Cherepovets districts - reached 5,65M by 1900AD [40][44]. Even faster growth occurred across Nizhny Novgorod (1,52M), Vyatka (3,03M), and Perm (2,99M), where state peasant land colonisation in Vyatka and the eighteenth- and nineteenth-century expansion of Ural iron and copper metallurgy multiplied the regional population more than eightfold from 915k in 1700AD to 2,42M in 1800AD and 7,65M in 1900AD [37][44]. In the Middle Volga (Kazan Guberniya at 2,17M in 1897, Simbirsk at 1,53M, Penza at 1,47M, and the Kasimov, Elatma, and Temnikov *uezds* at ~680k), multi-ethnic

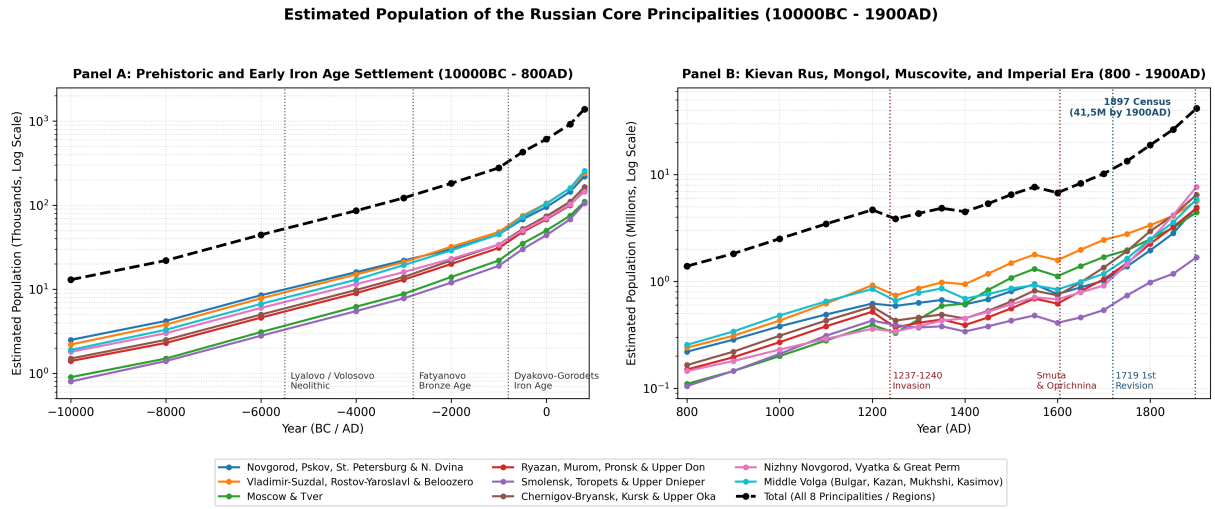


Figure 5. Estimated regional population trajectories across the eight historical regions of the Russian Core and Middle Volga-Ural basin (10000BC - 1900AD).

Russian, Tatar, Chuvash, Mordvin, and Mari agricultural growth raised the population from 1,18M in 1700AD to 2,47M in 1800AD and 5,85M in 1900AD [44][45].

Limitations

Three main sources of uncertainty affect pre-census population estimates across the Russian Core and Middle Volga-Ural regions. First, in the northern forest zone, shifting swidden cultivation (*podseka*) and periodic village relocation prior to the introduction of three-field rotation in the fourteenth and fifteenth centuries mean that unfortified rural *selishcha* were often occupied for only two to four generations, requiring contemporaneity adjustments when scaling from archaeological survey counts [20][24]. Second, sixteenth- and seventeenth-century Muscovite land cadastres (*pistsovye* and *perepisnye knigi*) assessed taxable peasant homesteads (*dvory*) and plough-units (*obzhi* and *sokhi*), providing incentives for extended families to cohabit within larger courtyard compounds and conceal untaxed dependents (*bobyli* and *zakhrebetniki*) prior to the introduction of the individual male soul tax in 1718-1724 [32][35]. Third, along the Middle Volga and Kama-Ural borderlands, non-Slavic tribute payers (*yasachnye lyudi*) were only partially incorporated into the early eighteenth-century revisions until the Third through Fifth Revisions (1762-1795) standardised enumeration across all fiscal estates [34][37].

Conclusion

Across the eleven millennia from the Younger Dryas to 1900AD, the population of the Russian Core and Middle Volga-Ural basin expanded from roughly 13k post-glacial foragers to 41,5M inhabitants on the eve of the twentieth century. Pre-Mongol growth was anchored by the urban trading network of Novgorod, Smolensk, Vladimir-Suzdal, Ryazan, and Volga Bulgaria, reaching 4,67M by 1200AD before the 1236-1240 Mongol campaigns reduced the regional total to 3,85M. Thereafter, internal colonisation of the Moscow-Tver and Upper Volga morainic uplands during the fourteenth and fifteenth centuries (6,49M by 1500AD), seventeenth-century peasant settlement south of the Oka abatis lines (10,19M by 1700AD), and eighteenth- and nineteenth-century black-earth agriculture and industrial urbanisation transformed the demographic balance between the old northern forest principalities and the southern and eastern

provinces.

Appendix

Population Estimates by Principality (10000BC - 1900AD)

Table 3 presents the population estimates (in thousands k and millions M) across all eight regional divisions from 10000BC to 1900AD.

Year	Novg/ SPb	Vlad-Suzd	Mosc-Tver	Ryaz-Don	Smol.	Oka-Kursk	N.Nov-Perm	Mid-Volga	Total
10000BC	2,5k	2,2k	0,9k	1,4k	0,8k	1,5k	1,8k	1,9k	13k
8000BC	4,2k	3,8k	1,5k	2,3k	1,4k	2,5k	3k	3,3k	22k
6000BC	8,5k	7,8k	3,1k	4,6k	2,8k	5k	6k	6,7k	44,5k
4000BC	16k	15k	6,2k	9k	5,5k	9,8k	11,5k	13k	86k
3000BC	22k	21k	8,8k	13k	7,8k	14k	16k	19,4k	122k
2000BC	30k	32k	14k	20k	12k	22k	23k	29k	182k
1000BC	45k	48k	22k	31k	19k	34k	34k	45k	278k
500BC	68k	75k	35k	48k	30k	52k	50k	72k	430k
1AD	95k	105k	50k	68k	44k	74k	70k	104k	610k
500AD	145k	160k	75k	100k	68k	110k	102k	160k	920k
800AD	220k	240k	110k	150k	105k	165k	145k	255k	1,39M
900AD	285k	310k	145k	195k	145k	220k	180k	340k	1,82M
1000AD	380k	430k	200k	270k	210k	310k	230k	480k	2,51M
1100AD	490k	620k	280k	380k	310k	430k	290k	650k	3,45M
1200AD	620k	920k	390k	520k	430k	580k	360k	850k	4,67M
1250AD	590k	740k	330k	370k	390k	430k	340k	660k	3,85M
1300AD	630k	860k	440k	410k	370k	460k	380k	780k	4,33M
1350AD	670k	980k	590k	440k	380k	490k	430k	860k	4,84M
1400AD	610k	940k	620k	390k	340k	450k	450k	690k	4,49M
1450AD	680k	1,18M	830k	460k	380k	530k	520k	760k	5,34M
1500AD	810k	1,49M	1,08M	560k	430k	650k	610k	860k	6,49M
1550AD	940k	1,78M	1,31M	690k	480k	820k	710k	920k	7,65M
1600AD	760k	1,58M	1,12M	620k	410k	740k	680k	840k	6,75M
1650AD	880k	1,98M	1,39M	810k	460k	980k	790k	990k	8,28M
1700AD	1,01M	2,45M	1,69M	1,05M	540k	1,35M	915k	1,18M	10,19M
1750AD	1,38M	2,78M	1,96M	1,48M	740k	1,92M	1,45M	1,64M	13,35M
1800AD	1,95M	3,35M	2,48M	2,25M	980k	2,95M	2,42M	2,47M	18,85M
1850AD	2,82M	4,12M	3,18M	3,24M	1,18M	4,15M	4,18M	3,58M	26,45M
1900AD	4,95M	5,65M	4,42M	4,85M	1,68M	6,45M	7,65M	5,85M	41,5M

Table 3. Regional population estimates across the eight historical divisions of the Russian Core and Middle Volga-Ural basin from 10000BC to 1900AD.

Appanage Population Estimates (1200 - 1450AD)

To map the eight regional divisions onto the individual fourteenth- and fifteenth-century appanage principalities (*udelnye knyazhestva*) and Golden Horde *uluses*, Table 4 provides population and density estimates at four medieval benchmarks (1200, 1300, 1380, and 1450AD).

Appanage Principality or Ulus	1200AD	1300AD	1380AD	1450AD	Core Density (inh./km², 1380)
Novgorod Republic	510k	505k	525k	530k	8,2 (Ilmen-Volkhov)
Pskov Republic	110k	125k	140k	150k	6,4 (Velikaya basin)
Vladimir, Pereyaslavl, Yuriev & Starodub	480k	410k	450k	540k	21,5 (Suzdal Opolye)
Rostov, Yaroslavl, Uglich, Mologa & Kostroma	330k	325k	375k	460k	9,8 (Rostov-Nero)
Beloozero & Vologda-Sheksna Appanages	110k	125k	145k	180k	2,4 (Sheksna-Beloye)
Grand Principality of Moscow, Mozhaysk, Vereya & Serpukhov	210k	240k	360k	545k	14,2 (Moskva-Radonezh)
Grand Principality of Tver, Kashin, Mikulin & Rzhev	180k	200k	245k	285k	11 (Upper Volga-Tvertsa)
Grand Principality of Ryazan, Pronsk & Yelets	410k	320k	335k	365k	7,8 (Middle Oka-Pronya)
Principality of Murom & Meshchera	110k	90k	90k	95k	4,5 (Lower Oka)
Principality of Smolensk, Vyazma & Toropets	430k	370k	365k	380k	5,2 (Upper Dnieper)
Principality of Bryansk, Karachev & Kursk-Posemye	340k	250k	255k	280k	5,8 (Desna-Sudost)
Upper Oka Appanages (Novosil, Odoev, Vorotynsk, Belyov, Tarusa, Mezetsk)	240k	210k	225k	250k	7,5 (Upper Oka-Upa-Zusha)
Nizhny Novgorod-Suzdal & Gorodets Principality	190k	205k	235k	270k	8,4 (Volga-Oka confluence)
Vyatka Land (Khlynov, Kotelnich, Orlov, Karino)	95k	100k	120k	155k	2,8 (Middle Vyatka)
Great Perm Principality (Cherdyn, Pokcha, Upper Kama)	75k	75k	85k	95k	1,1 (Upper Kama-Vishera)
Volga Bulgaria/Bulgar Ulus/ Kazan Khanate	680k	560k	570k	580k	14,5 (Bolgar-Kazan core)
Ulus of Mukhshi (Narovchat), Kasimov & Temnikov Principality	170k	220k	220k	180k	6,2 (Moksha-Tsna-Oka)

Table 4. Population and core settlement density estimates for medieval Russian appanage principalities and Golden Horde uluses (1200 - 1450AD).

Regional Density Estimates

Table 5 summarises the localised agro-ecological and urban settlement zones across the Russian Core and Middle Volga-Ural region from 10000BC to 1900AD, together with their population densities and primary sources.

Ecological and Settlement Zone	Date Range	Density (inh./km^2)	Area (km^2)	Ecological Zone & Historical/ Archaeological Basis
Butovo and Veretye Mesolithic	10000 - 5500BC	0,012	180k	Lacha-Vozhe-Sukhona & Volga-Oka peat-bog lacustrine necks (Nizhneye Veretye I, Stanovoye 4; Oshibkina 2006; Zhilin 2019, 2022)
Lyalovo and Volosovo Neolithic	5500 - 2500BC	0,065	240k	Upper/Middle Volga, Oka, Klyazma & Kama multi-hearth longhouse villages (60-200 m^2, 15-40 persons/house; Sakhtysh I-VIII; Kostyleva et al. 2014, 2022)
Fatyanovo and Abashevo Bronze Age	2800 - 1500BC	0,18	310k	Upper Volga-Moskva-Oka-Sura-Sviyaga pastoralist clan territories (>300 cemeteries led by Volosovo-Danilovsky; Nordqvist & Heyd 2020; Saag et al. 2021)
Dyakovo and Gorodets Iron Age	800BC - 750AD	0,55	420k	Moskva, Upper Volga, Oka, Desna & Kama bluff hillfort corridors (0,15-0,5 ha gorodishcha every 3-7 km, 50-150 persons/fort; Krenke 2011; Lopatina 2023)
Ilmen-Volkhov Poozerye Core	800 - 1900AD	14	6,5k	Lake Ilmen Poozerye, upper Volkhov (Rurikovo Gorodishche, Veliky Novgorod) & lower Shelona/Msta/Lovat pogost core (Nosov 1990; Shapiro et al. 1971)
Suzdal Opolye Loess Core	1050 - 1900AD	23	4,5k	Suzdal-Yuriev-Pereyaslavl-Vladimir dark-soil Opolye loess islands (400 medieval selishcha, 25-30% covering 5-19 ha; Modin et al. 2020; Shpolyanskiy 2021)
Moscow-Tver Watershed Hills	1280 - 1900AD	18,5	18k	Moscow-Radonezh-Zvenigorod-Dmitrov-Tver morainic watershed hills (three-field hamlet colonisation & Zamoskovny cadastres; Chernov 1991; Boytsov et al. 2020)
Upper Don Forest-Steppe	1150 - 1900AD	44,8	105k	Ryazan, Pronsk & Upper Don forest-steppe (medieval selishcha followed by 1719-1897 Tambov/ Voronezh/Ryazan agrarian boom; Vorotyntsev 2016; Myakisheva 2018)

(Continued from previous page)

Ecological and Settlement Zone	Date Range	Density (inh./km²)	Area (km²)	Ecological Zone & Historical/ Archaeological Basis
Bilyar-Bolgar Cheremshan Core	950 - 1400AD	18	16,5k	Western Zakamye (Bilyar 800 ha, Bolgar 380 ha, Suvar 95 ha, Cheremshan-Aktay chernozem agricultural heartland; Khuzin 2001; Rudenko 2012, 2018)
Kazan Predkamye Core	1380 - 1900AD	32,4	175k	Middle Volga (Kazan, Simbirsk, Penza & Kasimov/Temnikov from 1565-1568 cadastres to 1897 Census; Mustafina 2008; Abdullin et al. 2013; Troynitsky 1905)
Central Black Earth Belt	1719 - 1900AD	58,2	108k	Kursk, Orel, Tula & Kaluga forest-steppe and chernozem guberniyas from the 1st-10th Revisions to the 1897 Census (Kabuzan 1971; Rashin 1956; Troynitsky 1905)
Moscow and St. Petersburg Industrial	1861 - 1900AD	68,5	65k	Moscow & Saint Petersburg metropolitan industrial districts in the 1897 First All-Russian Imperial Census (Troitskaya & Ulyanova 2017; Rashin 1956; Troynitsky 1905)

Table 5. Population densities and inhabited areas across ecological and settlement zones of the Russian Core from 10000BC to 1900AD.

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